

Faculty Project Guide

Development of Cyber Competition Curriculum

Project Overview

Project Title: Cyber Competition Training Curriculum Development for Cyber Games

Project Duration: 6 Months

Project Lead: Vinay Kagitha (Mentor: Dr. John Goulden)

Department: Computer Science

Project Purpose

The purpose of this project is to design, develop, test, and document a cybersecurity training curriculum that prepares students for participation in individual and team-based cyber competitions, including the National Cyber League (NCL) and the National Centers of Academic Excellence (NCAE) Cyber Games.

The curriculum will begin with foundational knowledge and progress toward realistic competition scenarios. Rather than treating each competition category independently, instructional content will be organized around common cybersecurity knowledge domains shared across multiple competition formats. This approach reduces duplication while strengthening long-term technical competency and student confidence.

Project Goals

At the completion of this project, the faculty member will have developed a reusable instructional repository that includes:

- Complete workshop materials
- Hands-on lab exercises
- Instructor guides
- Student guides
- Presentation slides
- Virtual lab environments
- Practice competition exercises

Expected Project Outcomes

By the end of six months, the project should produce:

Curriculum

- Twelve complete instructional modules
- Fourteen-week workshop schedule
- Mini-workshop
- Competition preparation materials to include

Learning Resources

- Student workbooks with walkthroughs, exercises and challenges

- Instructor lesson plans
- PowerPoint presentations
- Lab manuals
- Challenge scenarios
- Assessment questions

Technical Resources

- Virtual machine templates
- Sample vulnerable environments
- Scripts
- Automation tools
- All resources to be stored so that it can be shared easily

Competition Resources

- Mock NCL exercises
- Mock NCAE Cyber Games scenarios
- Blue Team exercises
- Capture-the-Flag challenges
- Scoring rubrics
- Team readiness checklists

The faculty member is expected to research competition objectives and work on the deliverables listed. Faculty must work with their mentor and department Chair to get feedback and also plan on a workshop schedule to start hosting them for students.

Recommended Modules

Module 1 – Foundations	Module 2 – Defensive Operations	Module 3 – Networking
Topics • Linux Fundamentals • Windows Fundamentals • Networking Fundamentals • Virtualization • Git & Documentation Deliverables • Labs • Slides • Instructor guide • Student guide • Quiz/Assessment	Topics • Windows Hardening • Linux Hardening • Firewall Configuration • User Management • Secure Services • Patch Management • Logging • SIEM • Incident Response • Backup Deliverables • Blue Team labs • Hardening checklist • Labs • Slides • Instructor guide • Student guide • Quiz/Assessment	Topics • Packet Analysis • Wireshark • TCPDump • Network Mapping • IDS/IPS • Snort • Suricata Deliverables • Packet capture labs • Network troubleshooting exercises • Labs • Slides • Instructor guide • Student guide • Quiz/Assessment

Module 4 – Digital Forensics Topics • Event Logs • Memory Analysis • Registry • Browser Artifacts • Volatility • Autopsy Deliverables • Investigation labs • Case studies • Slides • Instructor guide • Student guide	Module 5 – Cryptography Topics • Classical Ciphers • AES • RSA • Hashing • JWT • Base64 Deliverables • Crypto challenge set • Slides • Instructor guide • Student guide	Module 6 – Password Security Topics • Hashcat • John the Ripper • Password Policies • MFA Deliverables • Password auditing lab • Slides • Instructor guide • Student guide
Module 7 – Web Security Topics • SQL Injection • XSS • CSRF • Burp Suite • OWASP Top 10 Deliverables • Web application labs • Slides • Instructor guide • Student guide	Module 8 – Reverse Engineering Topics • Assembly • Ghidra • GDB • Static Analysis • Dynamic Analysis Deliverables • Malware analysis exercises • Slides • Instructor guide • Student guide	Module 9 – Enumeration & Exploitation Topics • Nmap • SMB • LDAP • CVEs • Privilege Escalation • Metasploit Deliverables • Capture-the-Flag exercises • Slides • Instructor guide • Student guide
Module 10 – OSINT Topics • Google Dorking • WHOIS • DNS • GitHub Recon • Shodan • Maltego Deliverables • OSINT scavenger hunt • Slides • Instructor guide • Student guide	Module 11 – Capture the Flag Skills Topics • Methodology • Writeups • CyberChef • Pwntools • Automation Deliverables • Progressive CTF challenge repository • Slides • Instructor guide • Student guide	Module 12 – Competition Readiness Topics • Team Roles • Communication • Documentation • Stress Management • Scoreboard Strategy Deliverables • Mock competition • Team evaluation rubric • Slides • Instructor guide • Student guide

Recommended Six Month Timeline

Month 1 – Planning and Foundation

Objectives

- Finalize curriculum framework
- Build GitHub repository
- Configure lab environment
- Create templates
- Develop Module 1

Deliverables

- Project repository
- Module 1 complete
- Development standards
- VM baseline

Month 2 – Core Infrastructure

Objectives

- Develop Modules:
 - Module 2
 - Module 3
 - Module 4

Deliverables

- Blue Team labs
- Networking labs
- Forensics labs

Month 3 – Offensive Security

Content

Objectives

- Develop modules:
 - Module 5
 - Module 6
 - Module 7

Deliverables

- Crypto labs
- Password labs
- Web Security labs

Month 4 – Advanced Technical Topics

Objectives

- Develop modules:
 - Module 8
 - Module 9
 - Module 10

Deliverables

- Reverse Engineering labs
- Enumeration labs
- OSINT labs

Month 5 – Competition Development

Objectives

- Develop modules:
 - Module 11
 - Module 12
- Create:
 - Mock competitions
 - Scoring system
 - Team exercises

Deliverables

- Complete competition package

Month 6 – Validation and Deployment

Objectives

- Pilot the curriculum or go over the plan with the team, refine instructional materials, and prepare for adoption next semester.

Deliverables

- Pilot workshop implementation plan
- Student and faculty feedback summary forms
- Revised instructional materials
- Instructor handbook
- Final curriculum repository

Project Success Metrics

The project will be considered successful if the following criteria are achieved:

- All 12 instructional modules are completed.
- Every module includes slides, labs, instructor notes, and student guides.
- At least one hands-on assessment is created for each module.
- A complete 14-week workshop schedule is developed.
- A mock NCL-style competition is created.
- A mock NCAE Cyber Games scenario is developed.
- All lab environments are tested and documented.